Jiani (Elva) He

Data Science Consultant · Independent Researcher
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RESEARCH INTERESTS

Demand-forecast stability and planning value; causal decision support under partial information; trustworthy deployment of AI systems; AI-assisted reverse logistics.

This work corresponds to AI subfields on the U.S. Critical and Emerging Technologies List, including planning, reasoning, and decision making, and AI assurance and assessment techniques.

PROFESSIONAL EXPERIENCE

Accenture
Jun 2024 - Present
Data Science Consultant
Seattle, Washington

- Lead planning analyses, manage client relationships, and present recommendations to senior leaders.
- Analyze reverse-logistics cycle time and on-time delivery in SQL to identify bottlenecks; guide reuse, resale, recycling, or disposal decisions based on asset value and lifecycle stage.
- Analyze stock targets, available inventory, and demand; use an optimization engine to rebalance inventory across sites, address shortages, and maintain adequate supply.
- Plan quality-assurance component quantities using demand and bill-of-materials data; define service metrics and support automated inventory adjustments.
- Standardize prompts and analytical queries for enterprise LLM assistants; develop reusable guides and train analysts in AI-assisted analysis.

Accenture
Oct 2021 - May 2024
Data Science Senior Analyst
Seattle, Washington

- Built SQL dashboards and shortage alerts to compare inventory with demand and inform transfers between sites.
- Tracked reverse-logistics cycle times and on-time delivery; flagged bottlenecks and supported asset reuse, disposal, and recovery scheduling.
- Investigated inventory imbalances and abnormal cycle times using SQL and statistical analysis; automated reporting and purchase-order workflows.
- Across both Accenture roles, mentored five junior analysts in SQL, business problem solving, and client communication; led onboarding and team training.

BMW Group
Feb - Aug 2021
Data Scientist Intern · MIT Capstone, Dynamic Vehicle Pricing
Cambridge, Massachusetts

- Modeled vehicle demand and cross-price elasticity in Python to recommend discounts by customer segment and sales channel; built Tableau pricing dashboards.

EDUCATION

Massachusetts Institute of Technology
Aug 2021
MS, Business Analytics · Operations Research Center | GPA 4.7 / 5.0
Cambridge, Massachusetts

Classification trees for 40,000 news articles (Python/Julia) and robust bike-allocation models (Julia). Analytics Lab: an NLP admissions Q&A pipeline for AdmitHub with response evaluation and human review.

Purdue University
May 2020
BS, Industrial Engineering · Honors & Distinction | GPA 3.9 / 4.0
West Lafayette, Indiana

LSTM forecasting models (Keras) and reinforcement learning for sensor placement (Python/R).

RESEARCH & PUBLICATIONS

First-author research

1. **Jiani He**, Dingyan Shang, Yihua Xu, Shiqi Huang, Yan Lyu, Jize Li, Shangjing Tang. 2026. “Semantic Signal-Assisted Inspection and Recovery Allocation in Reverse Logistics.” *4th International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings 2026)*. arXiv:2609.02116. **Accepted · AIBThings 2026; proceedings forthcoming**

Accepted conference papers

2. Yihua Xu, Yan Lyu, Jize Li, **Jiani He**, You Lin, Yidan Hu, Rubing Qiang. 2026. “When Does an Added Planning Capability Create Operational Value? An Observable Evidence Standard.” *3rd World Congress on Industrial Engineering and Operations Management · Irvine, California*. arXiv:2610.27305. **Accepted · IEOM Irvine 2026; proceedings forthcoming**

3. Yan Lyu, Yihua Xu, **Jiani He**, Jize Li, Rubing Qiang, Yidan Hu. 2026. “From Weak Signals to Supply Chain Reliability Risk: A Feedback-Driven Risk Accumulation Framework for Late-Delivery Prediction, Early Warning, and Risk Prioritization.” *3rd World Congress on Industrial Engineering and Operations Management · Irvine, California*. Accepted manuscript · not yet publicly available. **Accepted · IEOM Irvine 2026; proceedings forthcoming**

4. Shiqi Huang, **Jiani He**, Dingyan Shang, Yihua Xu, Jize Li, Yan Lyu, Lashimi Muraleedharan Nair. 2026. “DACRI: Decision-Aware Causal Intervention Ranking for Critical Supply Chains.” *International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME 2026) · 15–17 October 2026 · Bali, Indonesia*. arXiv:2608.11154. **Accepted · ICECCME 2026; proceedings forthcoming**

5. Jize Li, **Jiani He**, Dishu Yang, Dingyan Shang, Jingjing Liu, Shiqi Huang. 2026. “Accuracy-Preserving Stability Regularization for Large-Scale Retail Demand Forecasting.” *ICEME 2026*. arXiv:2607.13331. **Accepted · ICEME 2026; proceedings forthcoming**

6. Yihua Xu, **Jiani He**, Ishita Chirag Talati, Yitian Qian, Youting Wang, Zhenyu Xu, Dingyan Shang. 2026. “Policy-Bounded, Provenance-Aware Deployment Assurance for LLM Request Routing.” *ICOAI 2026 · public preprint on SSRN*. SSRN 7554440. **Accepted · ICOAI 2026; proceedings forthcoming**

SERVICE & MENTORING

- Peer reviewer for 10+ journal manuscripts.
- Mentored five junior analysts across Accenture roles; led onboarding and team training, including AI-assisted analysis.

SKILLS

Tools: SQL, Python, R, Julia, Keras, Tableau, AWS

Modeling: Machine learning, Forecasting, Regression, Classification, Clustering, Robust optimization

Analytics & BI: Statistical analysis, Root-cause analysis, Data visualization, Dashboards, KPI design

AI: Generative AI, NLP, LLM workflows, Prompt & query design

Consulting: Problem framing, Client communication, Stakeholder coordination, Project delivery, Mentoring